



Pie charts

PIE CHARTS ARE A USEFUL VISUAL WAY TO PRESENT DATA.

A pie chart shows data as a circle divided into segments, or slices, with each slice representing a different part of the data.

Why use a pie chart?

Pie charts are often used to present data because they have an immediate visual impact. The size of each slice of the pie clearly shows the relative sizes of different groups of data, which makes the comparison of data quick and easy.



◁ Reading a pie chart

When a pie chart is divided into slices, it is easy to understand the information. It is clear in this example that the red section represents the largest group of data.

Identifying data

To get the information necessary to calculate the size, or angle, of each slice of a pie chart, a table of data known as a frequency table is created. This identifies the different groups of data, and shows both their size (frequency of data) and the size of all of the groups of data together (total frequency).

▽ Calculating the angles

To find the angle for each slice of the pie chart, take the information in the frequency table and use it in this formula.

$$\text{angle} = \frac{\text{frequency of data}}{\text{total frequency}} \times 360^\circ$$

For example:

$$\text{angle for United Kingdom} = \frac{375}{1,000} \times 360^\circ = 135^\circ$$

Annotations: "number of website hits" points to 375; "divide both numbers" points to the fraction bar; "total number of website hits" points to 1,000; "angle for pie chart" points to the result 135°.

The angles for the remaining slices are calculated in the same way, taking the data for each country from the frequency table and using the formula. The angles of all the slices of the pie should add up to 360°—the total number of degrees in a circle.

$$\text{United States} = \frac{250}{1,000} \times 360 = 90^\circ$$

$$\text{Australia} = \frac{125}{1,000} \times 360 = 45^\circ$$

$$\text{Canada} = \frac{50}{1,000} \times 360 = 18^\circ$$

$$\text{China} = \frac{50}{1,000} \times 360 = 18^\circ$$

$$\text{Unknown} = \frac{150}{1,000} \times 360 = 54^\circ$$

Country of origin	Frequency of data
United Kingdom	375
United States	250
Australia	125
Canada	50
China	50
Unknown	150
TOTAL FREQUENCY	1,000

◁ Frequency table

The table shows the number of hits on a website, split into the countries where they occurred.

"frequency of data" is broken down country by country

data from each country is used to calculate size of each slice

"total frequency" is total number of website hits from all countries

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◀ 204–205 Collecting and organizing data

◀ 206–209 Bar graphs

United Kingdom

135°